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First-principles calculations and thermodynamic modeling of the Yb-Ni binary system

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ABSTRACT

A complete thermodynamic description of the Yb-Ni binary system is developed by means of the CALculation of PHAse Diagram (CALPHAD) method in combination with first-principles calculations based on density functional theory (DFT) and available experimental data in the literature. Finite temperature thermodynamic properties of the Yb-Ni intermetallic compounds are predicted using the quasi-harmonic approach, where first-principles phonon calculations are performed to calculate the lattice vibrational entropy. The associate solution model is used to describe the complex thermodynamic behavior of the liquid phase. The calculated phase diagram agrees well the experimental phase equilibrium data in the literature. By the coupling of CALPHAD modeling with first-principles calculations, the present work provides a more thermodynamically accurate model of the Yb-Ni system when compared to previous models.

1. Introduction

In the last few decades, the intermetallic compounds between rare earth (RE) elements and transition metal elements have constituted a subject of intensive studies because of their appealing electronic, magnetic, optical and thermal properties. Among the RE-Ni systems, the Yb-Ni system may be of special interest for its extensive applications in magnetic, amorphous and catalytic materials [1–6]. The $YbNi_2$ and Yb_2Ni_2Al intermetallic compounds have attracted a great deal of attention for potential superconductivity associated to their intermediate valence, Kondo and heavy fermion behaviors [1,3]. The Al-Ni-Yb alloys, with the Al-rich precipitates and the Ni-Yb clusters coexisted, have been reported for great forming abilities of metallic glass and excellent mechanical properties [2,4]. In addition, Ni and Yb have been used as component elements for catalytic materials. For example, the Ni/Yb_2O_3 and $Yb\text{-}Ni/Si_2O$ catalyst deliver high conversion rate for the reaction of CH_4 oxidation and phenol hydrogenation, respectively [5,6]. However, the high vapor pressure of Yb can make the experimental synthesis of the Yb-Ni phases difficult. Therefore, considering for a systematic design and processing of the Yb-Ni-based materials, a comprehensive understanding of the thermochemistry and phase equilibria of the Yb-Ni system is of great importance.

Furthermore, Ni is a candidate device component for use in thermoelectric (TE) devices due to its electrical and thermal characteristics [7–9]. However, when the TE devices operate at high temperatures, the reactions between the Ni electrode and thermoelectric segments, and the diffusion of Ni into the segment may degrade the performance of the TE phase [7–9]. Among the current well-developed TE materials for high temperature (> 900 K) power generation, the complex Zintl compound, $Yb_{14}MnSb_{11}$, attracts a great deal of attention due to its substantially higher figure of merit and efficiency than any other competing materials [10]. Maintaining the stability between Ni and the $Yb_{14}MnSb_{11}$ phase is important in predicting the longevity of the TE device. This requires a comprehensive understanding of the phase stability and equilibria at the $Ni/Yb_{14}MnSb_{11}$ interface and can be assisted by a thermodynamic investigation of the Yb-Mn-Sb-Ni multi-component system. As one of the constituent binary systems, the Yb-Ni system contains several intermetallic phases which may form at the $Ni/Yb_{14}MnSb_{11}$ interface at high temperatures. The CALculation of PHAse Diagrams (CALPHAD) method has been proven to be an invaluable method to model the thermodynamic properties of the individual phases and predict the equilibria between the phases in a multi-component materials system [11]. The CALPHAD method establishes the thermodynamic models for all the phases in the unary, binary or higher-order systems to describe their Gibbs energy function, and optimizes the model parameters using the computational and experimental data

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Table 1
Crystallographic data of the intermetallic phases and pure elements in the Yb-Ni binary system together with details of first-principles calculations of each phase, including k-point mesh for electronic structure relaxation, supercell size, and k-point mesh for phonon calculations[a].

Phases	Pearson symbol	Space group	Prototype	Composition at% Yb	k-point mesh electronic[b]	Supercell size[c]	k-points mesh phonon
Ni (fcc)	cF4	$Fm\bar{3}m$	Cu	0.00	16×16×16	4×4×4	3×3×3
Yb_2Ni_{17}	hP38	P63/mmc	Th_2Ni_{17}	10.53	5×5×5	1×1×1	5×5×5
$YbNi_5$	hP6	P6/mmm	$CaCu_5$	16.67	9×9×10	2×2×2	3×3×3
Yb_2Ni_7	hR54	$R\bar{3}m$	Gd_2Co_7	22.22	9×9×1	N/A	N/A
$YbNi_3$	hR36	$R\bar{3}m$	$PuNi_3$	25.00	9×9×2	1×1×1	9×9×2
$YbNi_2$	cF24	$Fd\bar{3}m$	$MgCu_2$	33.33	6×6×6	2×2×2	3×3×3
α-YbNi	oP8	Pnma	FeB	50.00	7×11×9	2×2×2	3×3×3
β-YbNi	Unknown	Unknown	Unknown	50.00	N/A	N/A	N/A
Yb_3Ni	oP16	Pnma	Fe_3C	75.00	6×4×6	N/A	N/A
β-Yb (fcc)	cF4	$Fm\bar{3}m$	Cu	100.00	16×16×16	4×4×4	3×3×3
γ-Yb (bcc)	cI2	$Im\bar{3}m$	W	100.00	N/A	N/A	N/A

[a] The first-principles calculations were not performed for the β-YbNi and γ-Yb phases.
[b] The electronic k-point mesh is for the primitive unit cell of each phase.
[c] The supercell construction for phonon calculation is based on the primitive unit cell of each phase.

on phase equilibria and thermochemistry [11]. The model parameters are entered in a database and can be used for extrapolations to high-order systems.

In the present work, the thermodynamic data of the Yb-Ni intermetallic phases are predicted from first-principles calculations based on the density functional theory (DFT). Based on the results of the first-principles calculations and available experimental data [12–14], the Yb-Ni system is re-modeled using the CALPHAD method, and compared with the previous modeling results [15,16]. The constructed thermodynamic description of the Yb-Ni system will be used for the build of the thermodynamic database of the Mn-Yb-Sb-Ni multi-component system to study the interface stability between the Ni electrode and Yb-based thermoelectric materials in our future work.

2. Review of literature information

The experimental data on the thermochemistry and phase equilibria in the Yb-Ni system is rather scarce in the literature. The crystallographic information of the solid phases in the Yb-Ni system was first investigated by Buschow using X-ray diffraction (XRD) [17], as shown in Table 1. In addition to the intermetallic phases reported by Buschow, two other solid phases, namely Yb_2Ni_7 and Yb_3Ni, were synthesized under a high-pressure condition (7.7 GPa), which are possibly unstable at ambient pressure [18]. In 1973, Palenzona et al. constructed the Yb-Ni phase diagram in the whole composition range based on the measurements from differential thermal analysis (DTA) and metallography [12]. They also confirmed the structural information of the Yb-Ni intermetallic phases reported by Buschow [17]. They found that all the intermetallic phases are stoichiometric, and melt congruently or are part of a peritectic system. The thermodynamic behavior of the liquid phase is complicated, as reflected by the steep solid-liquid phase boundary in the composition range of $0.25 < x_{yb} < 0.5$ and the liquid miscibility gap at the Yb-rich side [12]. All of this implies that strong short-range ordering exists in the liquid phase at certain temperatures. The invariant reactions in the Yb-Ni system reported by Palenzona et al. are summarized in Table 2. Unfortunately, due to the presence of impurities in their Yb-starting materials, the invariant reactions and phase boundaries determined by Palenzona et al. may have some uncertainties [12]. For example, the melting temperature and bcc↔fcc polymorphic transformation temperature of Yb were reported to be 1098 K and 1022 K, respectively [12], whereas the accepted values for pure Yb are 1092 K and 1068 K [19].

Brutti et al. [13] measured the 298 K enthalpies of formation of five Yb-Ni intermetallic compounds using differential scanning calorimetry (DSC) and the Knudsen effusion (KE) method. However, the entropies of these intermetallic phases are still unknown, which can be critical to determine the stability of the compounds relative to the liquid phase. Most recently, Berezutskii et al. [14] measured the partial enthalpy of mixing of the Yb-Ni melts at 1300–1750 K using the isoperibolic calorimetry. Based on their measurement results, Berezutskii et al. were skeptical about the melting points of some Yb-Ni compounds and the steep liquidus in the composition range of $0.25 < x_{yb} < 0.5$, as reported by Palenzona et al.. They further postulated that the formation enthalpies of the $YbNi_2$ and α-YbNi compounds should be much smaller than the measured values by Brutti et al. [13] and proposed lower melting points of the $YbNi_3$ and $YbNi_2$ phases and a different phase boundaries between the liquid and the $YbNi_2$ phase [14]. However, their speculations have not been confirmed by any experiments.

Scarce experimental data with large uncertainties make the thermodynamic assessments of the Yb-Ni system intractable, though there are two reports in the literature [15,16]. Recently, Kardellass et al. [16] modeled the Yb-Ni system according to the phase equilibria data from Palenzona et al. [12] and liquid thermochemical data from Berezutskii et al. [14]. Their modeling results are in good agreement with the thermochemical data of the liquid phase. However, the calculated phase diagram shows large disagreement on the solid-liquid phase boundaries with the experimental data (marked as open circles) in the composition range of 16.7 ~ 50.0 at% Yb, as shown in Fig. 1a [16]. In addition, some of the invariant reactions predicted by Kardellass et al. [16] deviate significantly from the experimental results, as shown in Table 2. In fact, their modeling parameters were evaluated to fit the proposed phase boundaries by Berezutskii et al., which have never been confirmed by experiments [16]. Later on, another thermodynamic description for the Yb-Ni system was reported by Zhu et al. [15]. Their results are in fairly good agreement with most of the published experimental data on the phase equilibria and thermochemistry. However, the model failed to reproduce the invariant reaction of liquid ↔ $YbNi_2$ + α-YbNi and the congruent melting of the β-YbNi phase, both of which are supported by a dense set of experimental data points as shown in Fig. 1b. Furthermore, the Neumann-Kopp approximation was used for the modeling of the intermetallic phases, which yields artificial kinks in the heat capacity descriptions at the melting point of the constituent elements. Since the melting points of pure Yb is much lower than the Yb-Ni compounds, such a modeling strategy may lead to a less accurate and non-physical thermodynamic description of these compounds at high temperatures. Therefore, though two thermodynamic descriptions have been published recently, the further investigations on the thermodynamics of the Yb-Ni system are still needed.

First-principles calculations based on DFT have been demonstrated to be a reliable way of predicting missing thermochemical data for solid phases [11,20–22]. This approach enables the direct prediction of the structural and thermodynamic properties of the phase of interest by only using atomic numbers and crystal structure information as inputs [23,24]. In the present work, we performed first-principles quasi-

Table 2
Summary of the invariant reactions in the Yb-Ni system.

Reaction type	Reaction	$x(Yb)_{liq}$	T/K	Method	Reference
Eutectic	Liq ↔ fcc_Ni + Yb_2Ni_{17}	0.075	1503 ± 5	DTA/XRD/Metallograpy	[12]
		0.072	1458	Calculated	[16]
		0.085	1510	Calculated	[15]
		0.081	1510	Calculated	This work
Peritectic	Liq + $YbNi_5$ ↔ Yb_2Ni_{17}		1518 ± 5	DTA/XRD/Metallograpy	[12]
		0.107	1488	Calculated	[16]
		0.104	1518	Calculated	[15]
		0.097	1519	Calculated	This work
Peritectic	Liq + $YbNi_3$ ↔ $YbNi_5$		1568 ± 5	DTA/XRD/Metallograpy	[12]
Eutectic	Liq ↔ $YbNi_3$ + $YbNi_5$	0.209	1568	Calculated	[15]
		0.184	1566	Calculated	This work
Congruent	Liq ↔ $YbNi_3$		1583 ± 5	DTA/XRD/Metallograpy	[12]
			1583	Calculated	[15]
			1582	Calculated	This work
Peritectic	Liq + $YbNi_3$ ↔ $YbNi_2$		1528 ± 5	DTA/XRD/Metallograpy	[12]
		0.406	1330	Calculated	[16]
		0.343	1528	Calculated	[15]
		0.364	1525	Calculated	This work
Eutectic	Liq ↔ $YbNi_2$ + α-YbNi	0.455	1263 ± 5	DTA/XRD/Metallograpy	[12]
		0.457	1277	Calculated	This work
Congruent	Liq ↔ β-YbNi		1308 ± 5	DTA/XRD/Metallograpy	[12]
			1304	Calculated	This work
Polymorphic	α-YbNi↔ β-YbNi		1273 ± 5	DTA/XRD/Metallograpy	[12]
			1273	Calculated	[15]
			1278	Calculated	This work
Monotectic	Liq#1 ↔ β-YbNi + Liq#2	0.540/0.910	1298 ± 5	DTA/XRD/Metallograpy	[12]
		0.527/0.870	1303	Calculated	[16]
		0.568/0.937	1308	Calculated	[15]
		0.549/0.917	1296	Calculated	This work
Eutectic	Liq ↔ α-YbNi + bcc_Yb	0.981	1063 ± 5	DTA/XRD/Metallograpy	[12]
		0.986	1080	Calculated	[16]
		0.996	1088	Calculated	[15]
		0.989	1083	Calculated	This work
Eutectoid	bcc-Yb ↔ α-YbNi + fcc-Yb		1034	Calculated	This work

harmonic calculations to predict the finite temperature thermodynamic properties of the Yb-Ni intermetallic phases, including the enthalpy of formation, entropy, heat capacity and Gibbs energy. The predicted data enable a more accurate modeling of the Yb-Ni system and resolve the heat capacity artifact problem due to the Neumann-Kopp approximation. Furthermore, the calculated enthalpy of formation of the intermetallic compounds at 298 K are compared with the experimental data from Brutti et al. [13] and the previous modeling work [14,16].

3. Methodology

3.1. First-principles calculations for thermodynamic properties at finite temperatures

DFT-based first-principles calculations were performed to predict the finite temperature thermochemical properties of the solid phases that stable at the ambient pressure using the Vienna Ab-initio Simulation Package (VASP) [25]. In addition, the 0 K static energy of the two high-pressure stable compounds, Yb_2Ni_7 and Yb_3Ni, are also calculated via first-principles calculations. The calculation of the β-YbNi phase was not performed due to the lack of its crystal structure information. The projector augmented wave method (PAW) [25,26] and the exchange-correlation functional depicted by the generalized gradient approximation by Perdew, Burke, and Ernzerhof (PBE) [27] were employed. The energy cutoff of the plane wave was set as 350 eV to ensure the accuracy of the calculations. The pseudopotentials recommended by VASP were chosen for performing the present first-principles calculations. In particular, for element Yb, the electrons of the 5p6s shells are treated as the valence electrons, corresponding to the pseudopotential labeled as Yb_2; for Ni, the 4s3d shells are treated as the valence states. The energy convergence criterion of the electronic self-consistency was set as 10^{-7} eV/atom for all of the calculations. The reciprocal-space energy integration was performed by the Methfessel–Paxton method [28] for structural relaxations and phonon calculations. The phonon calculations were carried out using the supercell method. The Force constants, i.e. the Hessian matrix, were calculated directly using the VASP code [25]. The phonon density of states (DOS) and dispersion properties were predicted using the YPHON code based on a parameter-free, mixed-space approach [29,30]. The phonon calculations were not performed for the Yb_2Ni_7 and Yb_3Ni phases since they are not dynamically stable at zero pressure. The k-points grid and simulation supercell size used for structural relaxations and phonon calculations for each solid phase are provided in Table 1.

Based on the quasi-harmonic approach, the Helmholtz energy of a stoichiometric solid phase at temperature T and volume V consists of three terms [24],

$$F(V, T) = E_0(V) + F_{vib}(V, T) + F_{ele}(V, T) \quad (1)$$

The first term, $E_0(V)$, is the 0 K static energy without the contribution of the zero-point vibration. This term can be determined by a fitting of the first-principles energy vs. volume data points based on a four-parameter Birch-Murnaghan equation of state (EOS) [24],

$$E(V) = a + bV^{-2/3} + cV^{-4/3} + dV^{-2} \quad (2)$$

where a, b, c and d are fitting parameters. The minimum value of the fitting function is the 0 K static energy at the equilibrium state, and the corresponding volume is the equilibrium volume (V_0) at 0 K. In the present work, the data points for the EOS fitting were obtained by performing first-principle calculations at seven fixed volumes. A volume relaxation of the input supercell was first performed to obtain a start volume, V′. Then, the seven different volumes for the EOS fitting were chosen based on V′, namely 0.871V′, 0.913V′, 0.956V′, 1.000V′, 1.046V′, 1.093V′, 1.141V′, which are corresponding to an incremental change of 1.5% in the length of the three lattice parameters of the

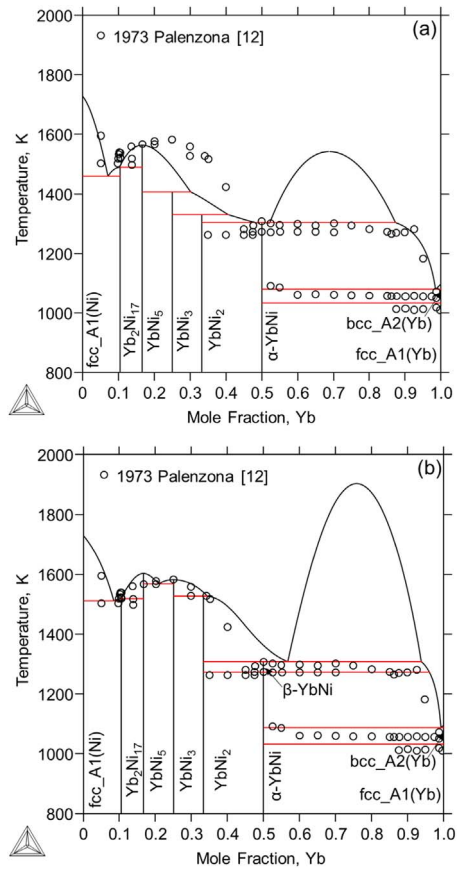

Fig. 1. Calculated Yb-Ni phase diagrams based on the previous thermodynamic assessments: (a) thermodynamic assessment of Kardellass [16]; (b) thermodynamic assessment of Zhu et al. [15]. Experimental data (○) by Palenzona et al. [12] are also included for comparison.

simulation supercell. The second term in Eq. (1), $F_{vib}(V,T)$, represents the contribution of lattice vibrations to the Helmholtz energy and is determinable by the phonon DOS under the quasi-harmonic approximation [21,24],

$$F_{vib}(V,T) = k_B T \int_0^{\infty} \ln\left[2\sinh\frac{\hbar\omega}{2k_B T}\right] g(\omega, V)d\omega \tag{3}$$

where $\hbar$ is the reduced Plank constant, ω the phonon frequency, and $g(\omega, V)$ denotes the phonon DOS at the frequency ω and volume V. The phonon DOSs of each phase of interest are calculated for the same seven volumes as those for the EOS fitting, which cover the temperature range from 0 K to the melting point of the phase of interest. The last term in Eq. (1), $F_{ele}(V,T)$, is the thermal electronic contribution to the Helmholtz energy, which is calculated based on the electronic DOS obtained from first-principles calculations [24]. For the Yb-Ni system, this part is necessary because the electronic densities of the intermetallic compounds at the Fermi level may not be zero. The $F_{ele}(V,T)$ term can be written as [24],

$$F_{ele} = E_{ele} - TS_{ele} \tag{4}$$

where E_{ele} and S_{ele} are the energy and entropy of thermal electron excitations, respectively. The entropy term can be calculated via integrating the electronic density of state (DOS) along with the distribution of the states over the energies of the electrons:

$$S_{ele}(V,T) = -k_B \int n(\varepsilon, V)[\ln f(\varepsilon, T) + (1 - f(\varepsilon, T))\ln(1 - f(\varepsilon, T))]d\varepsilon \tag{5}$$

where $n(\varepsilon, V)$ is the electronic density of state at energy ε and f is the Fermi-Dirac distribution since electrons are fermions, and given by:

$$f(\varepsilon, T) = \left[\exp\left(\frac{\varepsilon - \mu}{k_B T}\right) + 1\right]^{-1} \tag{6}$$

where μ is the chemical potential of electrons. The energy term can be calculated by:

$$E_{ele}(V,T) = \int_{-\infty}^{\infty} n(\varepsilon, V) f(\varepsilon, T)\varepsilon d\varepsilon - \int_{-\infty}^{\varepsilon_f} n(\varepsilon, V)\varepsilon d\varepsilon \tag{7}$$

where ε_f is Fermi energy.

In the present work, the obtained Helmholtz energy at zero pressure is taken as the Gibbs energy due to the negligible ambient pressure used in the current CALPHAD modeling. Other thermodynamic quantities at the equilibrium volume (V_0), such as heat capacity (C_P), entropy (S) and enthalpy (H), can be derived from the obtained Gibbs energy as a function of temperature based on basic thermodynamic relationships.

3.2. CALPHAD modeling

In the present work, nine phases, i.e., liquid, fcc_A1, bcc_A2, Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$, α-YbNi and β-YbNi, are modeled based on available experimental data from the literature and the thermochemical data from the first-principles calculations. The Gibbs energy functions of pure Yb and Ni for the CALPHAD modeling are taken from the Scientific Group Thermodata Europe (SGTE) pure element database [19], as implemented in the PURE5 database in Thermo-Calc software [31]. All the intermetallic phases are treated stoichiometrically. The Gibbs energy of the Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$ and α-YbNi phases above room temperature are described as,

$$G^{Yb_xNi_y} - H^{SER} = a + bT + cT\ln T + dT^2 + eT^{-1} + fT^3 \tag{8}$$

where a, b, c, d, e and f are the model parameters evaluated from the finite-temperature thermochemical data calculated by the first-principles quasi-harmonic method described above. H^{SER} is the stable element reference (SER) state that refers to the enthalpies of fcc-Yb and fcc-Ni at 298.15 K and 1 bar. Since the first-principles calculations were not available for the β-YbNi phase, its Gibbs energy expression is written as follow,

$$G^{\beta-YbNi} = 0.5\,{}^0G_{Yb}^{fcc} + 0.5\,{}^0G_{Ni}^{fcc} + A + BT + CT\ln T \tag{9}$$

where G_{Yb}^{fcc} and G_{Ni}^{fcc} are the Gibbs energy functions of pure fcc-Yb and fcc-Ni, respectively, taken from the SGTE compilation [19], and A, B and C are the model parameters to be evaluated, representing its Gibbs energy of formation.

To describe the steep liquidus in the small composition range of $0.25 < x_{Yb} < 0.5$ related to the strong short-range ordering, the associate solution model [21,32,33] was used to model the liquid phase in the present work. It is found that two associate species, i.e. $YbNi_3$ and YbNi, are necessary to describe the thermodynamic properties of the liquid phase. The Gibbs energy function of the liquid phase is expressed as,

$$G_m^{Liq}(T, y_i) = \sum_i y_i\,{}^0G_i^{Liq}(T) + RT\sum_i y_i \ln y_i + {}^{xs}G_L \tag{10}$$

where y_i is the mole fraction of specie i in the liquid phase, and ${}^{xs}G_L$ the excess-Gibbs-energy term. In Eq. (10), the terms ${}^0G_{Ni}^{Liq}$ and ${}^0G_{Yb}^{Liq}$ represent the Gibbs energy of pure Ni and Yb liquid, respectively, which are also taken from the SGTE compilation [19]. The Gibbs energies of the two associate species (i.e., $YbNi_3$ and YbNi) are described as,

$${}^0G_{YbNi_x}^{Liq} = \frac{1}{1+x}\,{}^0G_{Yb}^{Liq} + \frac{x}{1+x}\,{}^0G_{Ni}^{Liq} + {}^{YbNi_x}A + {}^{YbNi_x}BT + {}^{YbNi_x}CT\ln T \tag{11}$$

where ${}^{YbNi_x}A$, ${}^{YbNi_x}B$ and ${}^{YbNi_x}C$ are the model parameters for the Gibbs energy of formation of the associate species relative to pure Ni and Yb liquid phases. In, Eq. (10), the excess-Gibbs-energy term ${}^{xs}G_L$ is modeled with a Redlich-Kister polynomial [34],

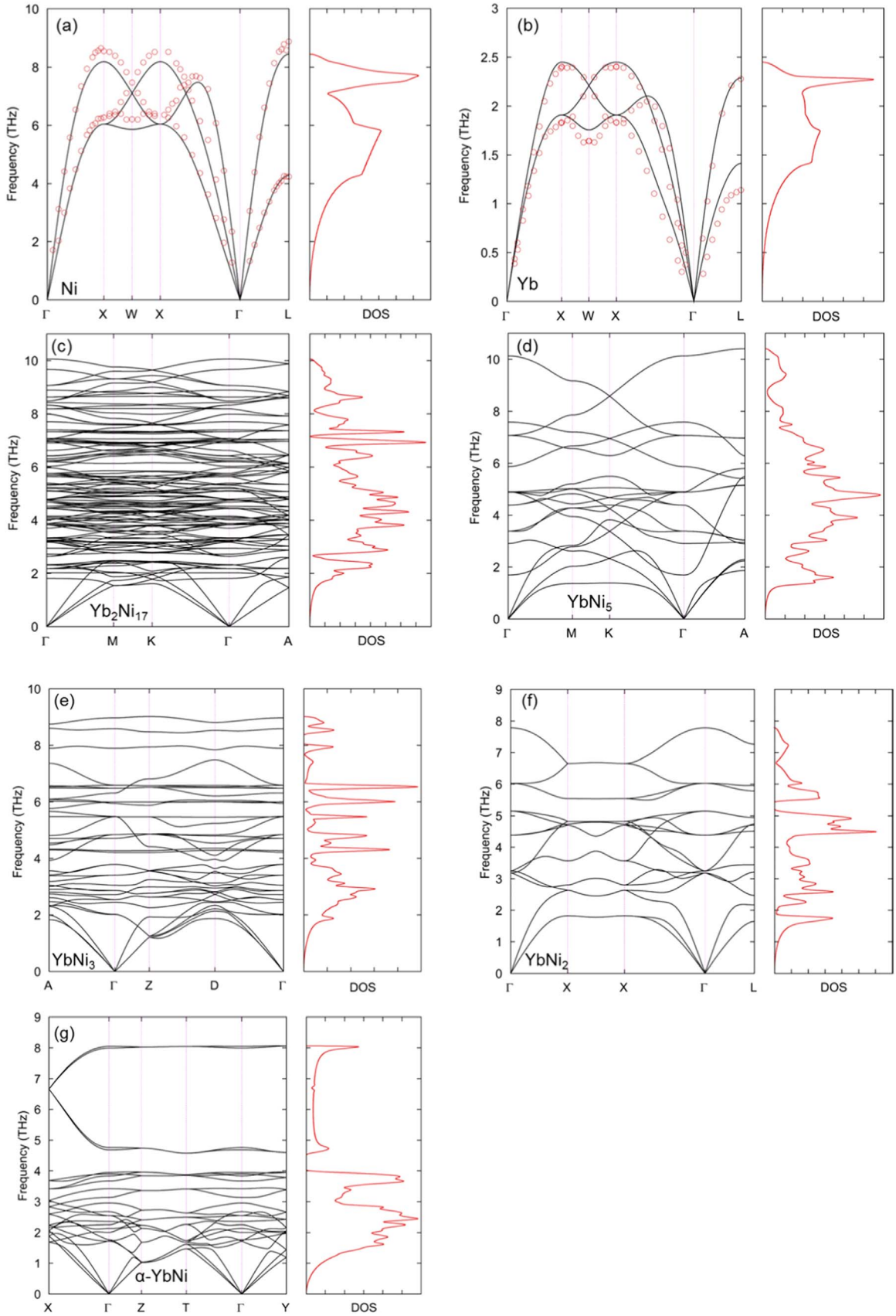

Fig. 2. Phonon dispersion curves and density of states (DOS) of the pure elements and intermetallic compounds in the Yb-Ni system: (a) fcc_Ni; (b) fcc_Yb; (c) Yb_2Ni_{17}; (d)$YbNi_5$; (e) $YbNi_3$; (f) $YbNi_2$; (g) α-YbNi. Available experimental data are also included for comparison (open circles) [38,39].

Table 3
First-principles predictions of lattice parameters, enthalpies of formation, entropies of formation, zero point vibrational energies of the intermetallic compounds in the Yb-Ni system at 0K (marked by *) and ambient conditions, and their Standard Element Reference states, fcc-Yb and fcc-Ni. Available experimental data [12,13,17] are also included for comparison. The experimental data are all measured at ambient conditions.

Phase	Method	Lattice parameter (Å)			Reference	Method	ΔH_f (kJ/atom-mol)	Reference	ΔS_f (J/atom-mol/K)	zero point vibrational energy (kJ/atom-mol)
		a	b	c						
fcc-Ni	XRD	3.524			[17]		-		-	
	XRD	3.525			[12]		-		-	
	FP PBE	3.531*			This work		-		-	3.602*
		3.540								
fcc-Yb	XRD	5.486			[17]		-		-	
	XRD	5.484			[12]		-		-	
	FP PBE	5.451*			This work		-		-	1.037*
		5.491								
Yb_2Ni_{17}	XRD	8.280		8.024	[17]	KE method	-13.9 ± 0.3	[13]		
	XRD	8.272		8.029	[12]					
	FP PBE	8.302*		8.043*	This work	FP PBE	-12.4*	This work	0.02	3.198*
		8.324		8.064			-12.4			
$YbNi_5$	XRD	4.847		3.962	[17]	KE method	-20.9 ± 4.4	[13]		
	XRD	4.842		3.959	[12]					
	FP PBE	4.879*		3.947*	This work	FP PBE	-21.0*	This work	0.84	2.955*
		4.899		3.963			-20.9			
Yb_2Ni_7	FP PBE	4.894*		35.790*			-22.1*	This work		
$YbNi_3$	XRD	4.920		24.160	[17]	KE method	-26 ± 4.8	[13]		
	XRD	4.919		24.140	[12]	DSC	-30 to -32	[13]	-0.86	
	FP PBE	4.931*		24.148*	This work	FP PBE	-30.9*	This work		2.767*
		4.945		24.217			-30.7			
$YbNi_2$	XRD	7.094			[17]	KE method	-32 ± 4.6	[13]		
	XRD	7.099			[12]	DSC	-30.0	[13]		
	FP PBE	7.080*			This work	FP PBE	-37.7*	This work	-0.79	2.543*
		7.101					-37.6			
α-YbNi	XRD	6.938	4.083	5.389	[17]	DSC	-28 ± 2.0	[13]		
	XRD	6.934	4.078	5.387	[12]					
	FP PBE	7.482*	4.064*	5.454*	This work	FP PBE	-27.4*	This work	1.31	1.918*
		7.506	4.077	5.472			-27.2			
Yb_3Ni	FP PBE	7.397*	9.467*	6.388*			-12.7*	This work		

$$^{xs}G_L = \sum_i \sum_{j<i} y_i y_j \sum_{v=0}^{n} {}^{v}L_{i,j}^{Liq}(y_i - y_j)^v \tag{12}$$

where ${}^{v}L_{i,j}^{Liq}$ is the vth interaction parameters between species i and j , and is given by [35,36],

$$^{v}L_{i,j}^{Liq} = ({}^{v,Liq}A + {}^{v,Liq}BT)\cdot\exp\left(-\frac{T}{{}^{v.Liq}\tau}\right) \tag{13}$$

where the exponential term is used for avoiding the artificial high-temperature miscibility gap caused by the conventional linear model [35,36]. The model parameters, ${}^{v,Liq}A$, ${}^{v,Liq}B$ and ${}^{v,Liq}\tau$ along with the parameters in Eq. (11) are evaluated based on the experimental thermochemical and phase equilibria data related to the liquid phase.

The Gibbs energies of the solid solution phases (G_m^{ϕ}) are described by the disordered substitutional solution model as given by,

$$G_m^{\varphi} = \sum_i x_i {}^{0}G_i^{\varphi} + RT\sum_i x_i \ln x_i + {}^{xs}G_{\varphi} \tag{14}$$

where R is the gas constant, x_i the mole fraction of i = Yb and Ni, ${}^{0}G_i^{\varphi}$ the Gibbs energy of pure Yb and Ni in the structure φ, also taken from the SGTE lattice stability database [19], and ${}^{xs}G_{\varphi}$ the excess Gibbs energy. The first and second term represent the pure element contributions and ideal mixing between Yb and Ni, respectively. The non-ideal interaction between Yb and Ni is captured by the excess Gibbs energy term, which is also modeled with a Redlich-Kister polynomial [34],

$$^{xs}G_{\varphi} = x_{Ni}x_{Yb}\sum_{k=0} {}^{k}L_{Ni,Yb}^{\varphi}(x_{Ni} - x_{Yb})^k \tag{15}$$

where ${}^{k}L_{Ni,Yb}^{\varphi}$ is the kth interaction parameter between Ni and Yb in the phase φ, which is also modeled in a A + BT expression. In the present work, we modeled the fcc_A1 and bcc_A2 solid solution phases.

The thermodynamic modeling of the Yb-Ni system was performed using the PARROT module of the Thermo-Calc software [31]. The evaluation of the model parameters started with the intermetallic phases Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$ and α-YbNi using the first-principles thermochemical data. For a given intermetallic phase, the parameters c, d, e and f in Eq. (8) are evaluated first using the calculated heat capacity values since heat capacity is directly related to the second derivative of the Gibbs energy with respect to temperature at a fixed pressure. It should be noted extrapolating Eq. (8) to high temperatures could result in an unrealistically high value of the heat capacity, which may artificially make the modeled phase becoming stable again. Therefore, in the present work, the heat capacity curve of each compound is cut off at a temperature that is about 50 K above its melting temperature [21,37]. Above the cutoff temperature, a constant C_P value is assumed due to unavailability of data.

Then, when the parameters related to heat capacity are fixed, the parameter b was determined by the 298 K entropy predicted by first-principles calculations, as it is the only remaining parameter that has a linear dependence to temperature. Afterwards the calculated enthalpy at 298 K relative to the SER state was used to evaluate the parameter a in Eq. (8). Once the Gibbs energy expressions of these intermetallic compounds were determined, the model parameters of the solid solution phases and the liquid phase were evaluated based on the phase equilibria data from Palenzona et al. [12] and thermochemical measurements from Berezutskii et al. [14]. In addition, the model parameters of the β-YbNi phase were evaluated based on its melting and polymorphic transformation temperatures. It should be noted that some of the enthalpy and entropy parameters (a and b in Eq. (8)) evaluated from the first-principles thermochemical data also need to be further modified in order to reproduce the experimental phase equilibria data accurately.

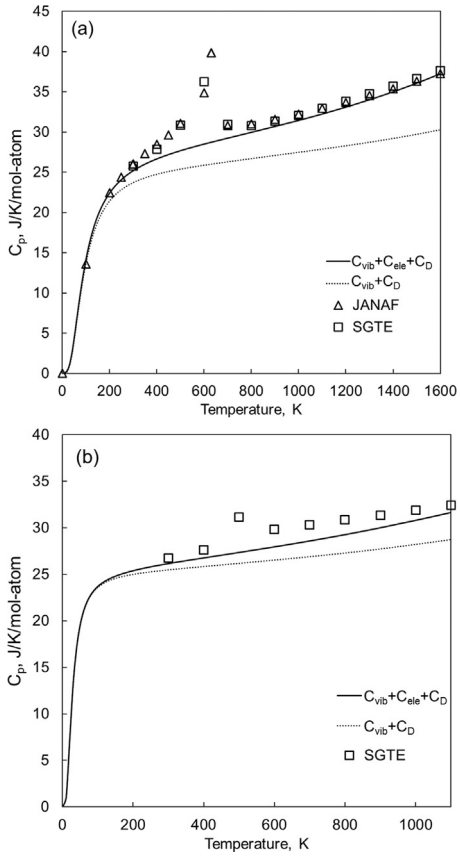

Fig. 3. DFT-predicted heat capacity of (a) fcc-Ni and (b) fcc-Yb with (solid lines) and without (dash lines) considering the thermal electronic contributions. Available experimental data are also included for comparison [19,40].

4. Results and discussion

4.1. Energetics of intermetallic phases by first-principles calculations

The calculated phonon dispersion curves of the fcc-Ni, fcc-Yb, Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$, and α-YbNi phases at the theoretical equilibrium volumes are shown in Fig. 2a-g, respectively, along with their phonon DOSs. To validate the phonon calculations in the present work, the calculated results of pure fcc Ni and fcc Yb are compared with the available experimental data [38,39] in Figs. 2a and 2b, showing good agreement. It is found that the maximum frequency of pure fcc Ni is much higher than that of pure fcc Yb, indicating stronger atomic bonding in fcc Ni. Consequently, the vibrational entropy of fcc Ni is lower than that of pure fcc Yb. From the agreement of the phonon dispersion curves for pure Ni and Yb, it is expected that the phonon properties of the Yb-Ni intermetallic phases can also be accurately predicted via first-principles calculations. As shown in Fig. 2c-f, imaginary phonon modes do not exist for all the compounds, indicating that they are all dynamically stable at low temperatures. It is found that the maximum frequency of the Yb-Ni intermetallic phase decreases with increasing Yb content so that the intermetallic phase with higher Yb content have a larger vibrational entropy.

To provide a benchmark for the first-principles methodology used in the present work, the calculated lattice parameters and enthalpies of formation at 0 K and 298 K of the pure elements and the intermetallic compounds are compared with available experimental data [12,17] in Table 3. The finite temperature lattice parameters are calculated based on the 0 K theoretical values by taking the effect of thermal expansion into account. As shown in Table 3, the predicted lattice parameters show remarkable agreement with the available experimental data [12,17]. The differences between the predictions and experiments are within 1% for most phases, while a relatively large discrepancy is observed for the *a* lattice parameter of the α-YbNi phase, which needs to be further investigated in the future work.

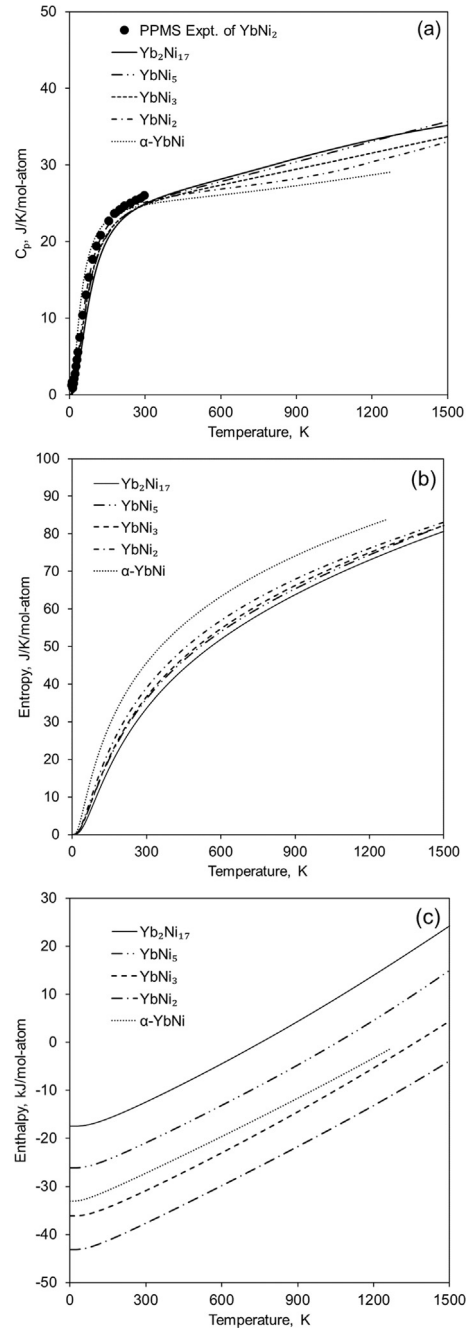

Fig. 4. Predicted thermodynamic properties of the intermetallic compounds in the Yb-Ni system from first-principles calculations: (a) Heat capacities in comparison with available experimental data [3]; (b) entropies; (c) enthalpies with respect to Standard Element Reference states.

Based on Eq. (1), the thermodynamic properties of a stoichiometric solid phase can be predicted via the quasi-harmonic phonon approach. The calculated heat capacity (C_P) of fcc-Ni and fcc-Yb are shown as a function of temperature in Figs. 3a and 3b, and compared with the experimental data from the SGTE database [19] and JANAF Thermochemical Tables [40], showing in a good agreement. The calculated heat capacities (C_P), entropies (S) and enthalpies (H) of the Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$ and α-YbNi phases are shown as a function of temperature in the Figs. 4a, 4b and 4c, respectively. The enthalpy values are reported relative to the SER state. As shown in Fig. 4, the

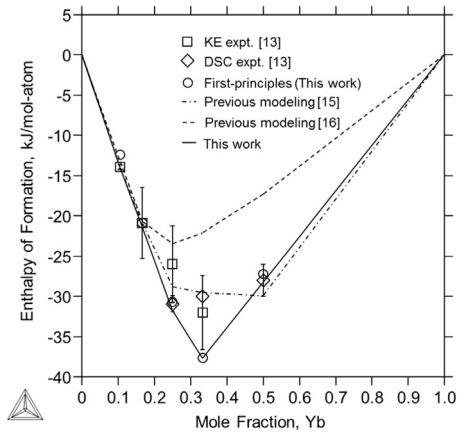

Fig. 5. Enthalpies of formation of the intermetallic compounds in the Yb-Ni system at 298 K from the present first-principles quasi-harmonic phonon calculations (○) and CALPHAD modeling (solid line), and in comparison with the experimental data [13] (KE method: □, and DSC method: ◊) and the previous CALPHAD modeling results (Zhu et. al. [15]: dot-dash line, and Kardellass et. al. [16]: dash line).

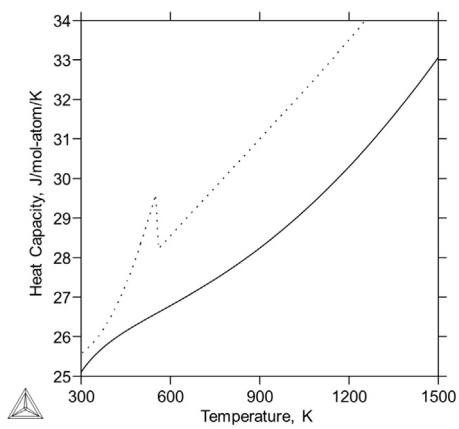

Fig. 6. Heat capacity of $YbNi_2$ from the present CALPHAD modeling (solid line) in comparison with that obtained from the Neumann-Kopp approximation (dashed line). The kink corresponds to the melting of bcc-Yb.

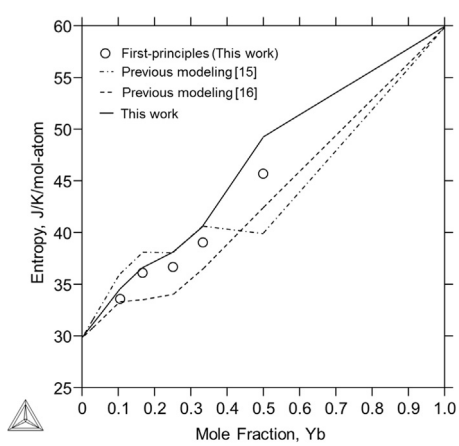

Fig. 7. Absolute entropies of the intermetallic compounds in the Yb-Ni system at 298 K from the present first-principles quasi-harmonic phonon calculations (○) and CALPHAD modeling (solid line), and in comparison with the previous CALPHAD modeling results (Zhu et al. [15]: dot-dash line, and Kardellass et al. [16]: dash line).

calculated heat capacity of the $YbNi_2$ phase agrees well with the measurements by Rojas et. al. from 4.2 K to 300 K [3]. For the convenience of usage, the predicted thermodynamic quantities shown in Fig. 4 are also listed as a function of temperature in Supplementary information (Table S1).

The theoretical enthalpy of formation of the intermetallic compounds listed in Table 3 is determined from,

$$\Delta_f H^{Yb_xNi_y} = H^{Yb_xNi_y} - \frac{x}{x+y}H^{fcc}_{Yb} - \frac{y}{x+y}H^{fcc}_{Ni} \tag{16}$$

where $H^{Yb_xNi_y}$, H^{fcc}_{Yb} and H^{fcc}_{Ni} are the predicted enthalpy per mole of atoms of the Yb_xNi_y, fcc-Yb and fcc-Ni phases from the first-principles calculations, respectively. Based on Eq. (16), the enthalpy of formation of the α-YbNi phase at 298 K is calculated to be −27.2 kJ/mol-atom, a value which agrees well with the recent experimental results from DSC measurements (−28 ± 2.0 kJ/mol-atom) by Brutti et al. [13]. Using the α-YbNi phase as a reference, Brutti et al. also derived the 298 K enthalpies of formation of the rest Yb-Ni intermetallic phases from the vaporization decomposition enthalpies measured by the Knudesn effusion (KE) method [13]. The enthalpies of formation of the Yb_2Ni_{17} and $YbNi_5$ phases determined by the first-principles calculations are −12.4 kJ/mol-atom and −20.9 kJ/mol-atom, respectively, which are also consistent with the values (−13.9 ± 0.3 kJ/mol-atom for Yb_2Ni_{17} and −20.9 ± 4.4 kJ/mol-atom for $YbNi_5$) obtained by the KE measurements [13]. Two preliminary DSC measurements on $YbNi_3$ and $YbNi_2$ have also been mentioned in the work by Brutti et al. [13], where the enthalpies of formation at 298 K were reported to be around −30 to −32 kJ/mol-atom for $YbNi_3$ and −30 kJ/mol-atom for $YbNi_2$ tentatively, which slightly differ from their KE results. The enthalpy of formation of the $YbNi_3$ phase predicted by the present calculations is −30.7 kJ/mol-atom, which is in good agreement with the DSC result, and within the uncertainty range of the value that obtained by the KE method [13]. The enthalpy of formation of the $YbNi_2$ phase at 298 K is predicted to be −37.6 kJ/mol-atom, which is more negative than the experimental values reported by Brutti et al. [13], but very close to the lower bound of the measurement uncertainty. The agreement between theoretical predictions in the present work and experimental measurements indicates that the enthalpies of formation data given by Brutti et al. [13] are reliable. In addition, the enthalpy of formation of Yb_2Ni_7 and Yb_3Ni phases are found to be above the convex hull, indicating they are indeed not stable at 0 K and 0 GPa pressure.

4.2. Thermodynamic modeling and phase equilibria

The enthalpy of formation of the Yb-Ni intermetallic phases at 298 K from the current CALPHAD modeling and first-principles calculations are shown in Fig. 5, along with the available experimental data [13] and previous CALPHAD modeling results [15,16]. The evaluation of the model parameters of these intermetallic phases in the present work was mainly based on the thermodynamic data from first-principles calculations. Since there are no eutectoid or peritectiod reactions reported between the Yb-Ni intermetallic phases at low temperatures [17], the modeling parameters for the Yb_2Ni_{17} and α-YbNi phases need to be slightly modified from the DFT-predicted values to bring their formation enthalpies on the convex hull. As shown in Fig. 5, the calculated enthalpy of formation from current CALPHAD modeling results are in good agreement with the first-principles predictions and the experimental data within the experimental uncertainties. In addition, the enthalpy of formation from the modeling work by Zhu et al. [15] are in fairly good agreement with experimental values, while the calculations from the modeling work by Kardellass et al. [16] deviate largely from both the experimental and first-principles calculations results.

In the previous thermodynamic models evaluated by Zhu et al. [15] and Kardellass et al. [16], the Neumann-Kopp approximation were used for the modeling of the intermetallic phases, where a $A+BT$ term were used to represent the Gibbs energy of formation. The parameters A and B are both assumed to be temperature independent. A typical modeling strategy is to evaluate the parameter A in terms of the enthalpy of formation at 298 K, and then adjust the parameter B according to the experimental phase equilibria data at high temperatures [15]. As

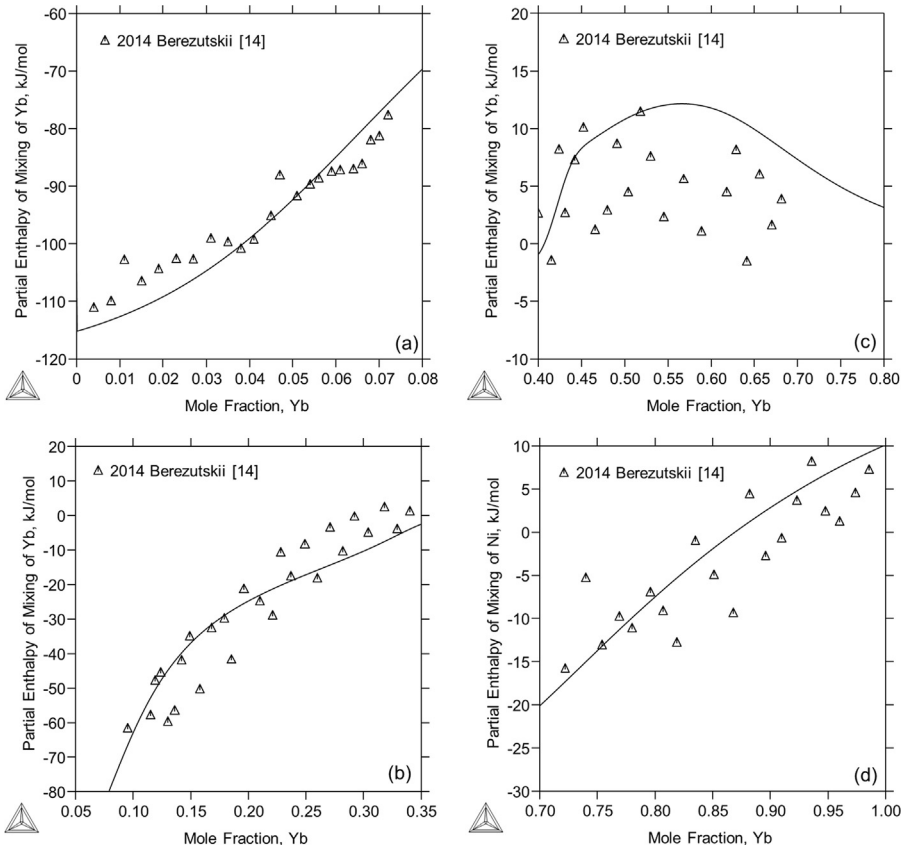

Fig. 8. Calculated partial enthalpy of mixing of Yb/Ni in the liquid phase in comparison with experimental data [14]: (a) Partial enthalpy of mixing of Yb at 1750 K; (b) partial enthalpy of mixing of Yb at 1600 K; (c) partial enthalpy of mixing of Yb at 1480 K; (d) partial enthalpy of mixing of Ni at 1300 K.

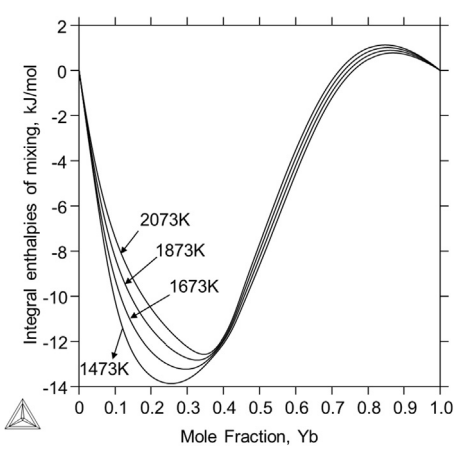

Fig. 9. Calculated integral enthalpies of mixing of the Yb-Ni liquid at different temperatures.

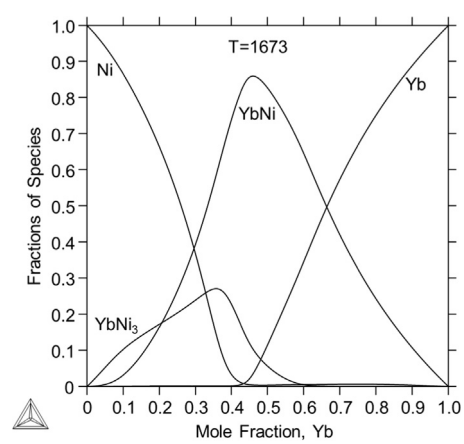

Fig. 10. Mole fractions of species in the Yb-Ni liquid solution as a function of total Yb composition at 1673 K.

discussed above, for the compound with melting points much higher than its constituent elements, the Neumann-Kopp approximation introduces a kink in the heat capacity of the compound at the melting temperature of the pure element. This is because the heat capacity of solid of a pure element is assumed to approach that of its liquid above melting temperature so the solid would not become stable at very high temperatures. Using the $YbNi_2$ phase as an example, Fig. 6 shows the comparison between the heat capacity calculated by the first-principles quasi-harmonic approach and the Neumann-Kopp approximation. It is shown that the Neumann-Kopp approximation introduces a kink at the melting temperature of pure Yb. This approximation on the heat capacity may result in a less accurate evaluation of the entropy parameter (i.e. parameter *B*). Fig. 7 shows the absolute entropies of the intermetallic compounds at 298 K calculated from the current CALPHAD modeling and the previous modeling works, [15,16] in comparison with the predictions from first-principles calculations. The current modeling results are in a fairly good agreement with the first-principles calculations. The small discrepancies are due to the further modifications on the entropy parameter in order to reproduce the experimental phase equilibria data. However, the entropy values calculated from the previous thermodynamic models [15,16] are less satisfactory in comparison with the first-principles results.

The model parameters of the liquid phase are evaluated from the available experimental thermochemical [14] and phase equilibria data [12]. An associate solution model, with the $YbNi_3$ and YbNi phases as the associates, is used to describe the possible short-range ordering in the liquid phase. Fig. 8a-d show the calculated partial enthalpies of dissolution Yb in liquid Ni at 1750, 1600 and1480 K, and Ni in liquid Yb at 1300 K, respectively, in comparison with experimental data from Berezutskii [14]. Good agreement between the current modeling results

Table 4
Thermodynamic parameters of the Yb-Ni binary system (in SI units).

Phase and model	Model parameters					
Liquid, (Ni, Yb, YbNi, $YbNi_3$) associate solution model	$G^{Liq}_{YbNi_3} = {}^0G^{Liq}_{Yb} + 3{}^0G^{Liq}_{Ni} - 65153 + 7.191T$					
	$G^{Liq}_{YbNi} = {}^0G^{Liq}_{Yb} + {}^0G^{Liq}_{Ni} - 21096 + 2.100T - 4.315T \ln T$					
	${}^0L^{Liq}_{Ni,YbNi_3} = -73931 + 19.500T$					
	${}^1L^{Liq}_{Ni,YbNi_3} = 19061 - 25.500T$					
	${}^0L^{Liq}_{YbNi_3,YbNi} = (10149 - 89.610T)\cdot\exp(-\frac{T}{15822})$					
	${}^1L^{Liq}_{YbNi_3,YbNi} = (30310 - 20.036T)\cdot\exp(-\frac{T}{7064})$					
	${}^2L^{Liq}_{YbNi_3,YbNi} = (-10245 - 40.126T)\cdot\exp(-\frac{T}{14243})$					
	${}^0L^{Liq}_{YbNi,Yb} = 16105 + 3.100T$					
	${}^1L^{Liq}_{YbNi,Yb} = -17000 + 3.500T$					
	${}^2L^{Liq}_{YbNi,Yb} = -7000$					
fcc_A1 (Ni, Yb)	${}^0L^{fcc}_{Ni,Yb} = 10000$					
bcc_A2 (Ni, Yb)	${}^0L^{bcc}_{Ni,Yb} = 10000$					
Intermetallic compounds	$G^{Yb_xNi_y} - H^{SER} = a + bT + cT\ln(T) + dT^2 + eT^{-1} + fT^3$ (J/mol-atom)					
	a	b	c	d	e	f
Yb_2Ni_{17}						
$298 < T < 1570$	− 21,204.2	117.506	− 22.1817	-5.57766×10^{-3}	18,876.751	2.6960×10^{-7}
$1570 < T$	− 29,231.1	215.996	− 35.6930	0	0	0
$YbNi_5$						
$298 < T < 1620$	− 28,922.4	122.796	− 23.4054	-3.67400×10^{-3}	34,313.054	-9.7057×10^{-8}
$1620 < T$	− 40,130.3	222.593	− 36.8113	0	0	0
$YbNi_3$						
$298 < T < 1630$	− 39,179.7	124.102	− 23.8853	-2.94858×10^{-3}	36,567.120	-7.4965×10^{-8}
$1630 < T$	− 48,348.3	204.390	− 34.6652	0	0	0
$YbNi_2$						
$298 < T < 1580$	− 46,196.3	140.666	− 26.9936	1.08470×10^{-3}	72,664.360	-6.9618×10^{-7}
$1580 < T$	− 54,288.3	200.120	− 33.9354	0	0	0
α-YbNi						
$298 < T < 1330$	− 35,837.1	116.294	− 24.5661	-1.02569×10^{-3}	21,212.797	-1.9773×10^{-7}
$1330 < T$	− 39,448.4	151.854	− 29.3690	0	0	0
β-YbNi	$G^{\beta-YbNi} = 0.5{}^0G^{fcc}_{Yb} + 0.5{}^0G^{fcc}_{Yb} - 18031.1 + 1.085T - 1.515T\ln T$					

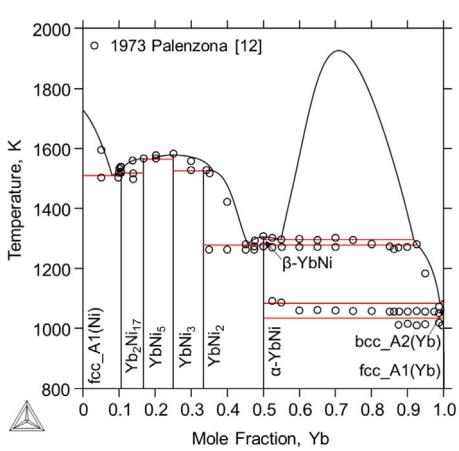

Fig. 11. Calculated phase diagram of the Yb-Ni system from the present thermodynamic modeling, and in comparison with experimental data of Palenzona et al. [12].

and experimental data can be found in Figs. 8a, 8b, and 8d. Although the partial enthalpies of mixing of Yb in liquid Ni at 1480 K from the current model is slightly higher than the experimental data as shown in Fig. 8c, the calculated value lies in a reasonable range considering the experimental scatter. The calculated integral enthalpies of mixing of the Yb-Ni liquid at various temperatures are plotted in Fig. 9. Fig. 10 shows the calculated species fractions in liquid as a function of Yb concentration at 1673 K.

The evaluated self-consistent thermodynamic parameters of the Yb-Ni system are summarized in Table 4 with the calculated temperature-composition phase diagram shown in Fig. 11. The experimentally measured liquidus temperatures and invariant reactions are well reproduced by the present thermodynamic model considering the experimental scatter. The relatively large discrepancies between the calculated results and the experimental measurements are seen for two invariant reactions. The temperature of the eutectic reaction, i.e., liquid ↔ bcc_Yb + α-YbNi is experimentally measured as 1063 K, while the calculated temperature is 1082 K. In fact, the experimentally measured temperature is even lower than the commonly acceptable polymorphic transformation temperature between fcc-Yb and bcc-Yb, which is 1065 K [41]. As discussed earlier, this is due to the presence of impurities in the Yb starting material. Since there is no solubility of Ni in bcc_Yb, the actual temperature of the eutectic reaction Liq ↔ bcc_Yb + α-YbNi should lie between 1068 K and 1092 K, as suggested by Nash [42]. The calculated eutectic temperature from the current thermodynamic model is within this temperature range. The predicted eutectic temperature of the liquid ↔ $YbNi_2$ + α-YbNi is about 14 K higher than the experimental measurements; however, it was not possible to accurately reproduce the other data in addition to this reaction without over-parameterizing. The issue related to over-parameterizing and the uncertainty on determining the phase boundary in the present work could be addressed in the future work using a newly developed approach, as implemented in the ESPEI/pycalphad package [43]. The liquid miscibility gap is predicted to be existed over 1773 K, in a good agreement with the metallographic observation [12]. It was observed that the Yb-Ni alloys with compositions in the range of the miscibility gap always have a two-layered microstructure after rapidly cooling

from 1773 K [12]. The two-layered microstructure was considered to result from the two immiscible liquids with different mass densities.

5. Conclusion

The experimental phase diagrams and thermodynamic data and thermodynamic models of the Yb-Ni system in the literature were critically reviewed. First-principles quasi-harmonic phonon calculations were performed to investigate the phonon density of state and dispersion properties of five stoichiometric compounds, Yb_2Ni_{17}, $YbNi_5$, $YbNi_3$, $YbNi_2$ and α-YbNi and predict their finite-temperature thermodynamic properties. The first-principles predictions provide robust input thermochemical data for the thermodynamic modeling of these phases and avoid the artifact of their heat capacity descriptions from the Neumann-Kopp approximation. The complexity of the liquid phase is described by an associate solution model. A complete set of self-consistent thermodynamic parameters are obtained via the CALPHAD method based on the available experimental data and the first-principles predictions. Comprehensive comparisons between the calculated and measured thermodynamic quantities indicate that the existing experimental information in the Yb-Ni system can be satisfactorily accounted for by the present thermodynamic description. The obtained thermodynamic databases will be used to build the Yb-Mn-Sb-Ni multicomponent database in an effort to understand interface stability between Ni and thermoelectric materials.

Acknowledgements

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2017.09.004.

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